

Figure 1: Max-cut: certified approximation ratio of random greedy (RG, mean over trials) with respect to each upper bound, as a function of the cardinality k . $f(S)$ = weight of edges crossing $(S, V \setminus S)$; $G(n, p)$ with integer weights in $\{1, \dots, 9\}$. RG/OPT uses the exact optimum (brute force) where it was computable. Additional baselines: the LP over the dual's base sets (B2), the pruned double-greedy bound γ_1 (B3) and the modular bounds μ_2, μ_3 on the pruned lattice (B4); B3 and B4 bound the unconstrained optimum. Marginal is $\min_S f(S) + \text{Pen}(S) + \sum_{a \in T_k(S)} [f(a \mid S)]^+$ ($T_k(S)$: the k largest marginals) over the same base sets, i.e. the dual bound without its caps. Hybrid (B5) is the LP B2 strengthened with the dual's prefix caps, and $\min(\text{dual}, B5)$ the smaller of the dual bound and B5. Curves are means over graph/sample seeds.

Random greedy: certified ratio vs. k , with the hybrid LP

Each plot shows RG/ B for every upper bound B of the baseline figures plus the hybrid LP ($B5 = B2$ with the dual's prefix caps) and $\min(\text{dual}, B5)$, against the $1/e$ guarantee of random greedy.

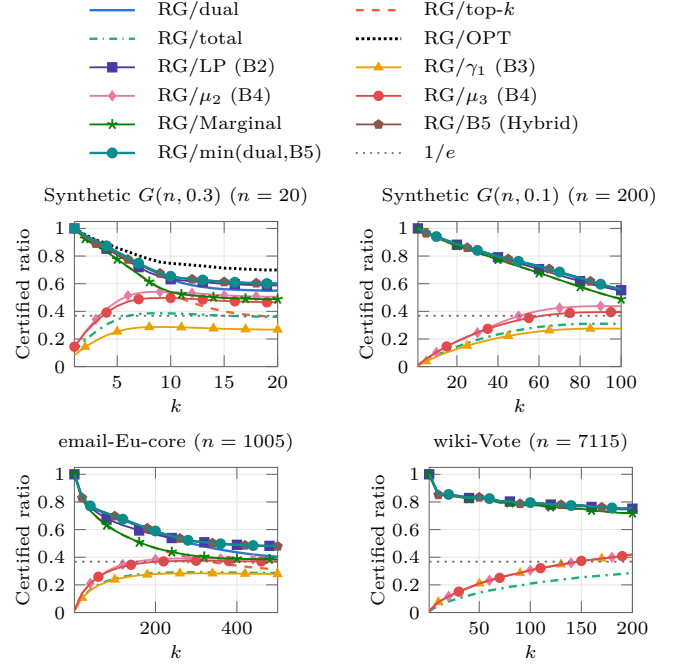


Figure 2: Directed cut: certified approximation ratio of random greedy (RG, mean over trials) with respect to each upper bound, as a function of the cardinality k . $f(S)$ = weight of edges leaving S ; synthetic $G(n, p)$ with random orientations and SNAP graphs. RG/OPT uses the exact optimum (brute force) where it was computable. On the largest instances the dual bound is minimized over every 10th greedy prefix only. Additional baselines: the LP over the dual's base sets (B2), the pruned double-greedy bound γ_1 (B3) and the modular bounds μ_2, μ_3 on the pruned lattice (B4); B3 and B4 bound the unconstrained optimum. Marginal is $\min_S f(S) + \text{Pen}(S) + \sum_{a \in T_k(S)} [f(a \mid S)]^+$ ($T_k(S)$: the k largest marginals) over the same base sets, i.e. the dual bound without its caps. Hybrid (B5) is the LP B2 strengthened with the dual's prefix caps, and $\min(\text{dual}, B5)$ the smaller of the dual bound and B5. Curves are means over graph/sample seeds.

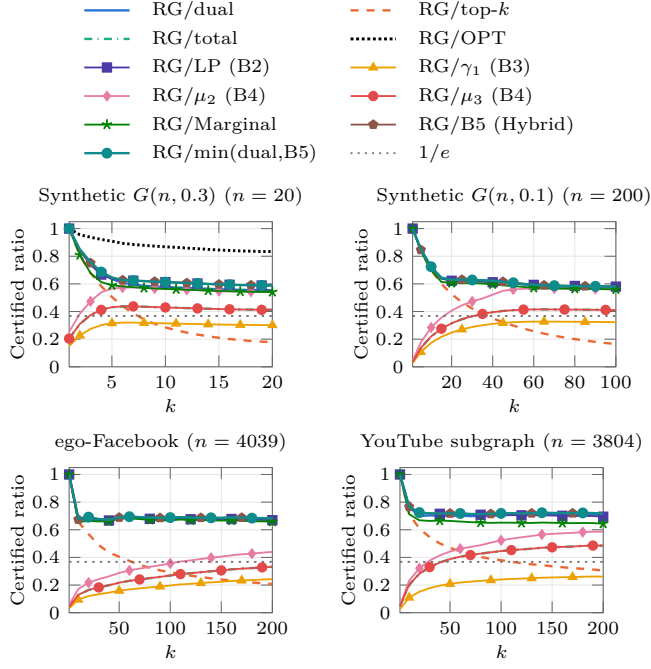


Figure 3: Revenue maximization: certified approximation ratio of random greedy (RG, mean over trials) with respect to each upper bound, as a function of the cardinality k . $f(S) = \sum_{i \notin S} \sqrt{\sum_{j \in S} w_{ij}}$ with $w_{ij} \sim U(0, 1)$. RG/OPT uses the exact optimum (brute force) where it was computable. On the largest instances the dual bound is minimized over every 5th or 10th greedy prefix only. Additional baselines: the LP over the dual’s base sets (B2), the pruned double-greedy bound γ_1 (B3) and the modular bounds μ_2, μ_3 on the pruned lattice (B4); B3 and B4 bound the unconstrained optimum. Marginal is $\min_S f(S) + \text{Pen}(S) + \sum_{a \in T_k(S)} [f(a \mid S)]^+ (T_k(S): \text{the } k \text{ largest marginals})$ over the same base sets, i.e. the dual bound without its caps. Hybrid (B5) is the LP B2 strengthened with the dual’s prefix caps, and min(dual, B5) the smaller of the dual bound and B5. Curves are means over graph/sample seeds.

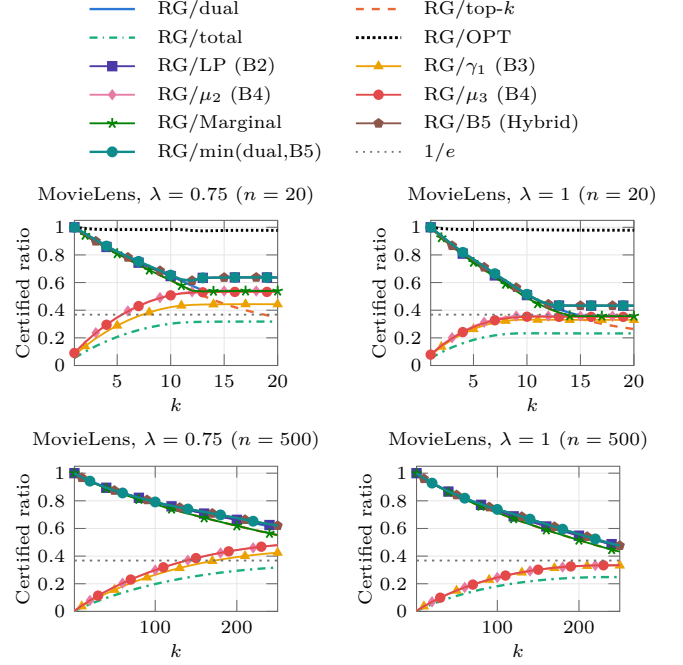


Figure 4: Diverse recommendation: certified approximation ratio of random greedy (RG, mean over trials) with respect to each upper bound, as a function of the cardinality k . $f(S) = \sum_{i \in S} \sum_{j \in V} s_{ij} - \lambda \sum_{i,j \in S} s_{ij}$, cosine similarities of MovieLens 1M rating vectors. RG/OPT uses the exact optimum (brute force) where it was computable. Additional baselines: the LP over the dual’s base sets (B2), the pruned double-greedy bound γ_1 (B3) and the modular bounds μ_2, μ_3 on the pruned lattice (B4); B3 and B4 bound the unconstrained optimum. Marginal is $\min_S f(S) + \text{Pen}(S) + \sum_{a \in T_k(S)} [f(a \mid S)]^+ (T_k(S): \text{the } k \text{ largest marginals})$ over the same base sets, i.e. the dual bound without its caps. Hybrid (B5) is the LP B2 strengthened with the dual’s prefix caps, and min(dual, B5) the smaller of the dual bound and B5. Curves are means over graph/sample seeds.

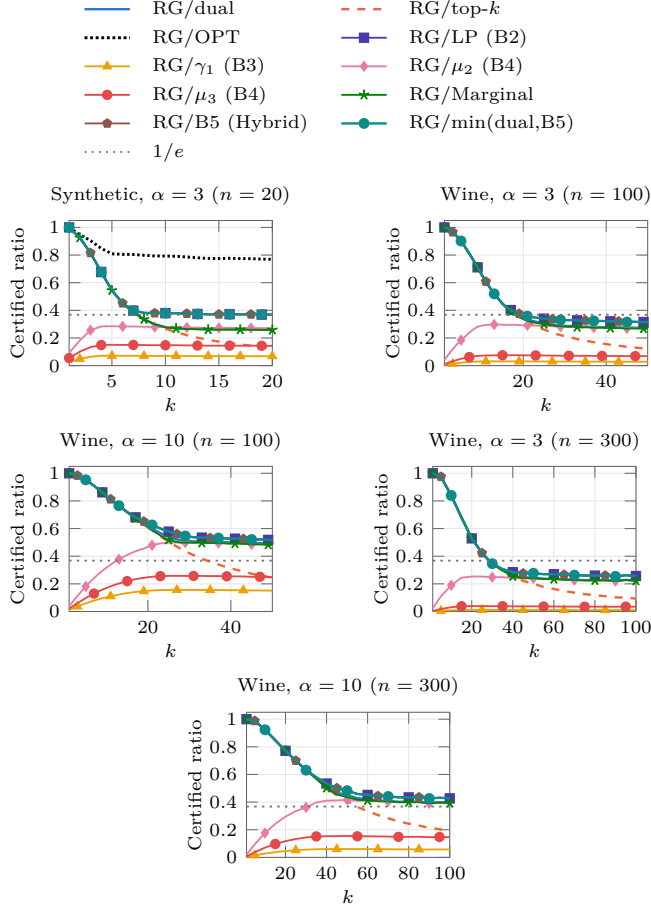


Figure 5: Log-determinant: certified approximation ratio of random greedy (RG, mean over trials) with respect to each upper bound, as a function of the cardinality k . $f(S) = \log \det(L_S)$, $L = \alpha(K + 10^{-3}I)$ with an RBF kernel K on the features. RG/OPT uses the exact optimum (brute force) where it was computable. On the largest instances the dual bound is minimized over every 5th greedy prefix only. Additional baselines: the LP over the dual’s base sets (B2), the pruned double-greedy bound γ_1 (B3) and the modular bounds μ_2, μ_3 on the pruned lattice (B4); B3 and B4 bound the unconstrained optimum. Marginal is $\min_S f(S) + \text{Pen}(S) + \sum_{a \in T_k(S)} [f(a | S)]^+$ ($T_k(S)$: the k largest marginals) over the same base sets, i.e. the dual bound without its caps. Hybrid (B5) is the LP B2 strengthened with the dual’s prefix caps, and min(dual, B5) the smaller of the dual bound and B5. Curves are means over graph/sample seeds.

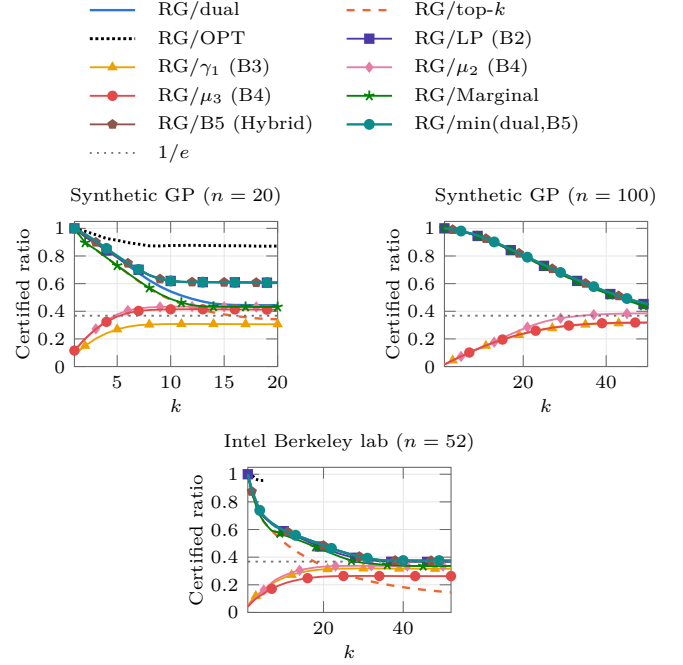


Figure 6: Gaussian mutual information: certified approximation ratio of random greedy (RG, mean over trials) with respect to each upper bound, as a function of the cardinality k . $f(S) = I(X_S; X_{V \setminus S})$ for a Gaussian process on random points and for the Intel Berkeley lab temperature sensors. RG/OPT uses the exact optimum (brute force) where it was computable. Additional baselines: the LP over the dual’s base sets (B2), the pruned double-greedy bound γ_1 (B3) and the modular bounds μ_2, μ_3 on the pruned lattice (B4); B3 and B4 bound the unconstrained optimum. Marginal is $\min_S f(S) + \text{Pen}(S) + \sum_{a \in T_k(S)} [f(a | S)]^+$ ($T_k(S)$: the k largest marginals) over the same base sets, i.e. the dual bound without its caps. Hybrid (B5) is the LP B2 strengthened with the dual’s prefix caps, and min(dual, B5) the smaller of the dual bound and B5. Curves are means over graph/sample seeds.

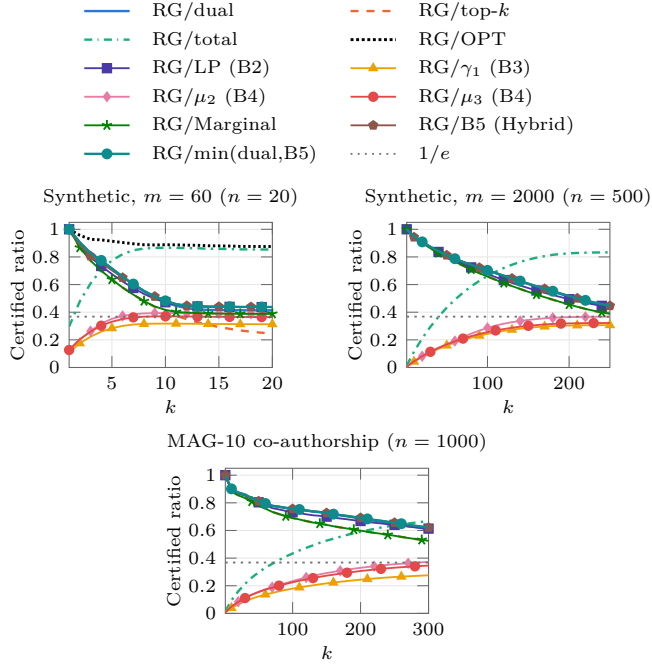


Figure 7: Hypergraph cut: certified approximation ratio of random greedy (RG, mean over trials) with respect to each upper bound, as a function of the cardinality k . $f(S)$ = weight of hyperedges split by S ; random hypergraphs and the MAG-10 co-authorship hypergraph. RG/OPT uses the exact optimum (brute force) where it was computable. Additional baselines: the LP over the dual’s base sets (B2), the pruned double-greedy bound γ_1 (B3) and the modular bounds μ_2 , μ_3 on the pruned lattice (B4); B3 and B4 bound the unconstrained optimum. Marginal is $\min_S f(S) + \text{Pen}(S) + \sum_{a \in T_k(S)} [f(a \mid S)]^+ (T_k(S):$ the k largest marginals) over the same base sets, i.e. the dual bound without its caps. Hybrid (B5) is the LP B2 strengthened with the dual’s prefix caps, and min(dual, B5) the smaller of the dual bound and B5. Curves are means over graph/sample seeds.